The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 January 2020.

INCH-POUND

MIL-PRF-19500/162F
4 September 2019
SUPERSEDING
MIL-PRF-19500/162E
2 September 2014

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICES, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N1614, 1N1615, 1N1616, 1N4458, AND 1N4459, NORMAL AND REVERSE POLARITY,
STUD MOUNT PACKAGE, QUALITY LEVELS JAN, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, power rectifier diodes. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package outline is a case mounted stud (TO-203AA, formerly DO-4) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types (1)	V_{RWM}	I_O		I_F	Barometric pressure (reduced)	I_{FSM} at: t = 1/120 s T_C = +150°C	T_J and T_{STG}
		T_C = +25°C (2)	T_C = +150°C				
	V (pk)	A dc	A dc	A dc	mm Hg	A	°C
1N1614, R	200	10	5	15	8	100	
1N1615, R	400	10	5	15	8	100	−65
1N1616, R	600	10	5	15	16	100	to
1N4458, R	800	10	5	15	30	100	+175
1N4459, R	1,000	10	5	15	54	100	

(1) Electrical characteristics for R suffix devices are identical to the corresponding non-R suffix devices unless otherwise specified.
(2) Derate linearly at 66.67 mA dc/°C above T_C = +25°C.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics. Primary electrical characteristics are not applicable for this specification sheet.

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "1614", "1615", "1616", "4458" and "4459".

1.5.3 Reverse polarity suffix symbol. The "R" suffix symbol is used to identify reverse polarity devices.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

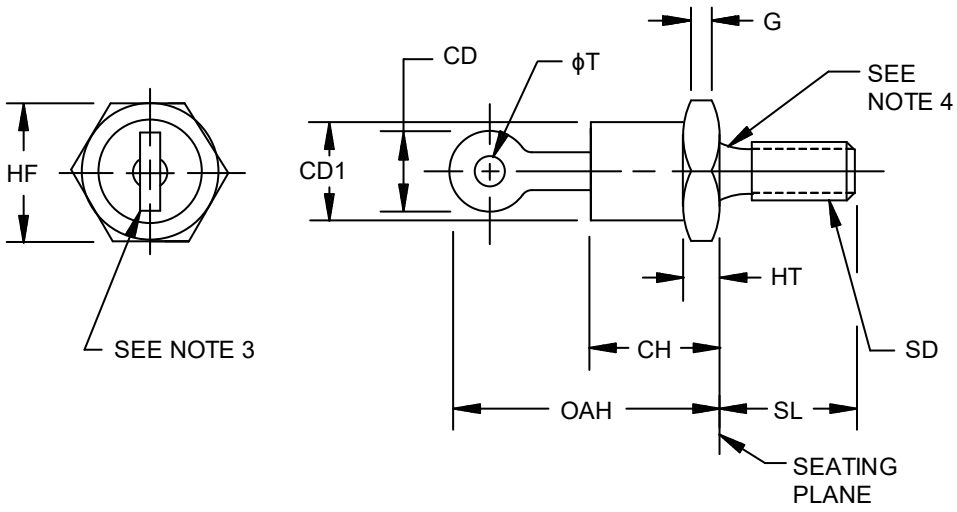
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD		.250		6.35	5
CD1		.424		10.77	
CH		.405		10.29	
G	.060		1.52		
HF	.424	.437	10.77	11.10	
HT	.075	.175	1.91	4.45	
OAH		.800		20.32	
SD					6, 7, 8
SL	.422	.453	10.72	11.51	
ϕT	.060		1.52		

NOTES:

1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. The ASME thread reference is 10−32 UNF−2A.
3. Angular orientation of this terminal is undefined.
4. Diameter of unthreaded portion .189 inch (4.80 mm) maximum and .163 inch (4.14 mm) minimum.
5. Terminal end shape is unrestricted.
6. The maximum pitch diameter of plated threads shall be basic pitch diameter .169 inch (4.29 mm).
7. Unit shall not be damaged by torque of 15 inch-pound (1.7 N-m) applied to 10−32 UNF−2B nut assembled on thread.
8. Complete threads shall extend to within .078 inch (1.98 mm) of the seating plane.
9. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for DO−203AA package.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements on MIL-PRF-19500 and this document.

3.4.3 Reverse and standard polarity types. Reverse and standard types are electrically identical. The standard types have the cathode connected to the stud and the reverse types have the anode connected to the stud.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. The polarity shall be indicated by an arrow graphic symbol with the arrow pointing to the cathode end for forward bias.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Tests in either polarity shall be sufficient to obtain qualification approval of both polarities.

4.2.1 Group E qualification. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

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4.3 Screening (quality level JANTX and JANTXV levels only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screening	Measurement
	JANTX and JANTXV levels
(1) 3b	Surge (see 4.3.1)
(1) 3c	Thermal impedance (see 4.3.2)
9	Not applicable
11	I_{R1} and V_F
12	See 4.3.3
13	Subgroup 2 of table I herein: ΔI_{R1} ≤ 100 percent of initial reading or ±5.0 µA dc, whichever is greater, ΔV_F ≤ ±0.1 V dc.

(1) Surge shall precede thermal impedance and shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

4.3.1 Surge current. The surge current test shall be performed in accordance with method 4066 of MIL-STD-750. The following details shall apply: I_O = 0; $V_{RM(W)}$ = 0; I_{FSM} = 100 A; six surges; T_A = room ambient as defined in the general requirements of 4.5 of MIL-STD-750. One surge per minute maximum. tp = 8.3 ms.

4.3.2 Thermal impedance ($Z_{\theta JX}$) measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum limit and conditions for $Z_{\theta JX}$ in screening shall be derived by each vendor by means of process control of actual measurements which characterizes the die attach process. When three lot date codes have exhibited control, the data from these three lots shall be used to establish a fixed screening limit (not to exceed the group A limit).

4.3.3 Power burn-in conditions. Power burn-in shall be performed in accordance with condition A of method 1038 of MIL-STD-750. The following details shall apply: T_A = +150°C; V_{RWM} = V_{RWM} (see 1.3); I_O = 0; t = 48 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table E-V of MIL-PRF-19500, and table I herein.

4.4.1.1 Thermal impedance measurements for group A. The following test conditions shall be used for $Z_{\theta JX}$ measurements in a group A, subgroup 2 inspection:

a. I_M measuring current 50 to 250 mA.

b. I_H forward heating current 3 to 10 A.

c. t_H heating time 150 to 400 ms.

d. t_{MD} measurement delay time 50 to 300 µs maximum.

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4.4.1.2 Thermal impedance limits for group A. The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIB (JANTX and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be I_{R1} and V_F in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
B2	4066	Condition A, I_{FSM} = 100 A (pk); ten surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 5 A dc; V_{RWM} = rated V_{RWM} (see 1.3); T_C = +150°C.
B3	1037	.25 rated $I_O \leq I_O$ applied ≤ rated I_O; 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end points) shall be I_{R1} and V_F in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A, 10 pounds (4.54 kg), t = 15 ±3 s.
C2	2036	Test condition D1, 10 ounce/inch (7.06 N-cm), t = 15 ±3 s.
C2	2036	Test condition F, method B, 5 pounds (2.27 kg), t = 15 ±3 s.
C2	2036	Test condition D2, method B, 15 pounds (6.8 kg), t = 15 ±3 s.
C6	1037	.25 rated $I_O \leq I_O$ applied ≤ rated I_O; 2,000 cycles.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be I_{R1} and V_F in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
E5	1001	Hg = rated Hg, t = 60s, (see 1.3), V_R = rated V_{RWM}.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection of conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature, T_A = +25°C ±3°C.

4.5.3 Thermal resistance. Thermal resistance measurement shall be performed in accordance with method 3101 or 4081 of MIL-STD-750. Forced moving air or draft shall not be permitted across the device during test.

4.5.4 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750, (read and record value $Z_{\theta JX}$). Derived conditions, limits, and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	Condition B, I_F = 15 A(pk), $t_p \leq 8.3$ ms, duty cycle ≤ 2 percent, pulsed (see 4.5.1)	V_F		1.5	V (pk)
Reverse current leakage	4016	DC method; V_R = rated V_{RWM} (see 1.3)	I_{R1}		50	µA dc
Thermal impedance 2/	3101	See 4.3.2	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
Reverse current leakage	4016	DC method; V_R = rated V_{RWM} (see 1.3), T_C = +150°C	I_{R2}		500	µA dc
Subgroups 4, 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point electrical measurements, this test is required for the following subgroups:
Group B, subgroups 2 and 3 (JAN and JANTX).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock	1056	100 cycles, 0°C to 100°C.	
Electrical measurements		See subgroup 2 of table I herein.	
Hermetic seal	1071		
Electrical measurements		See subgroup 2 of table I herein.	
Subgroup 2			45 devices, c = 0
Steady-state dc blocking life	1048	1,000 hours.	
Electrical measurements		See subgroup 2 of table I herein.	
Subgroup 4			
Thermal impedance curves		See subgroup 4 of table E-IX of MIL-PRF-19500.	
Subgroup 5			22 devices, c = 0
Barometric pressure, reduced (altitude operation)	1001	Hg = rated Hg (see 1.3), t = 60 s, While the test is being performed, I_R shall be monitored and shall not exceed table I limits.	
1N1614, R		400 V dc	
1N1615, R		400 V dc	
1N1616, R		600 V dc	
1N4458, R		800 V dc	
1N4459, R		1000 V dc	

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

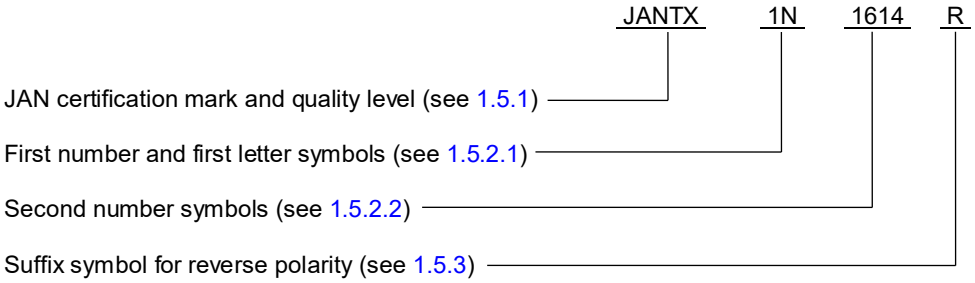
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices and are constructed using the following forms.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices and covered by this specification sheet.

Quality level JAN	Quality level JANTX	Quality level JAN	Quality level JANTX
Standard polarity		Reverse polarity	
JAN1N1614	JANTX1N1614	JAN1N1614R	JANTX1N1614R
JAN1N1615	JANTX1N1615	JAN1N1615R	JANTX1N1615R
JAN1N1616	JANTX1N1616	JAN1N1616R	JANTX1N1616R
JAN1N4458	JANTX1N4458	JAN1N4458R	JANTX1N4458R
JAN1N4459	JANTX1N4459	JAN1N4459R	JANTX1N4459R

6.6 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 692-6939 or DSN 850-6939.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army − CR
Navy − EC
Air Force − 85
DLA − CC

Review activities:
Army − AR, MI
Navy − AS, MC, SH
Air Force − 19

Preparing activity:
DLA − CC

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